

BioSpectral Systems

Visible Light Hygiene

The Impact of Circadian Rhythms on Health

The human body is a highly intelligent system, carrying out millions of physiological processes without our conscious input. From hormone production to mitochondrial energy transfer and water generation, these processes don't occur at random; they are deeply influenced by cues from our environment. One of the most powerful influences on these biological processes is light, both visible and invisible, alongside temperature changes.

One of the best-known cycles governing our biological processes is the **Circadian Rhythm**. This cycle plays a critical role in various bodily functions, influencing sleep, metabolism, gene expression, mental clarity, body composition, detoxification, and more. Circadian rhythms are biological clocks that operate on a roughly 24-hour cycle, aligning our internal processes with light and dark. This synchronization, known as **chronobiology**, is crucial for optimal health.

Nearly every tissue and organ in the body has its own biological clock, regulated by a master clock in the brain. This master clock, the **suprachiasmatic nucleus (SCN)** in the hypothalamus, keeps all the body's biological clocks in sync, responding primarily to signals from the eyes and the light they detect. Humans, being diurnal, are naturally aligned to be active during the day and rest at night. Our biology is tethered to the cycles of light and dark set by the sun, making the timing of light exposure essential for optimal function.

Artificial Light and Circadian Disruption

Modern lighting, particularly LED lights, has introduced an unnatural spike of blue light, unseen in natural sunlight. This artificial light, especially when used after sunset, disrupts the natural cues our bodies rely on, confusing our biological clocks and often leading to poor sleep, impaired healing, and other health challenges. Being indoors and on devices more than ever before means that our biological systems are increasingly out of sync with natural light cycles. This disruption in circadian rhythm impacts many areas of health, from metabolism and digestion to hormone production. With the transition from the warmer, more eye-friendly incandescent lighting to the harsher LED lighting, it has become important to actively manage our light exposure to maintain circadian health.

How Circadian Rhythm Drives Health and Performance

Life on Earth is driven by energy, primarily from sunlight. Sunlight not only powers life but also dictates the rhythm of countless biological processes. Evolution has equipped every organism with internal clocks, known as **circadian rhythms**, which anticipate and adapt to the presence and absence of sunlight. The term "circadian" is derived from Latin, meaning "approximately a day," because these cycles align with a 24-hour rhythm.

Nearly every process in the body, from cellular energy production to hormone secretion, follows a circadian rhythm or is directly influenced by it. For instance, mitochondria, the cellular engines, rely on sunlight for optimal function, transferring energy from food into usable biological

energy. As a result, our digestion, stress hormone secretion, and even repair mechanisms are timed with the sun's presence.

Here are some keyways circadian rhythms affect our physiology:

- **Energy Production:** Sunlight via its interaction with melanin, biological water, hemoglobin in red blood cells and mitochondria is critical for powering oxidative phosphorylation and proton and electron currents within the body including the vital DC electric current.
- **Mental Concentration:** A few hours after sunrise as the ultraviolet light from the sun begins to directly reach our skin and eyes from the sun, this kicks up our neurotransmitters especially dopamine which sparks our mental alertness, executive function and focus and present consciousness power. Digestion.
- **Hormones:** Hormone production, like cortisol, thyroid hormone and testosterone peak in the morning, while dopamine and leptin peak during solar noon, and serotonin and melatonin at night. These and all other hormones follows a hourly, daily or multi-day rhythm to support wakefulness, sleep, reproduction, growth and so much more.

When our circadian rhythm is disrupted, whether by artificial light, poorly timed meals, or stress, our body feels the impact, often resulting in fatigue, digestive issues, and other symptoms.

Restoring Light Hygiene for Optimal Health

To combat the pervasive influence of artificial lighting, we can take steps to align our light exposure more closely with natural patterns:

- **Morning Sunlight:** Begin the day with natural sunlight exposure to set your biological clocks for wakefulness and energy production.
- **Limit Blue Light at Night:** Minimize exposure to artificial blue light after sunset by using warmer lighting or blue-blocking technology.
- **Eat According to Sunlight:** Align meals with daylight hours to support optimal digestion and metabolic function.

By being mindful of light exposure and other environmental cues, we can restore harmony to our internal clocks, ensuring our bodies function as they were evolutionarily designed to—with the sun as our guide.

Bright artificial light after sunset, especially from LEDs, screens, and TVs, sends a false daytime signal to the brain. This suppresses melatonin, the hormone that tells the body it is time to sleep and begin overnight cellular repair.

A 2022 clinical study found that even moderate light exposure during sleep, around 100 lux like a bedside lamp, led to higher heart rates and reduced insulin sensitivity by the next morning.

Instead of resting and restoring, the body stayed in a stress response state throughout the night.

<https://www.nhlbi.nih.gov/news/2022/study-sleeping-even-small-amount-light-could-harm-heart-health>

Another study showed that people exposed to typical LED lighting before bed burned less fat during sleep than those exposed to light with reduced blue wavelengths. Sleep is when the body usually switches to fat burning for energy, and blue light blocks that switch.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8196130/>

In a study tracking more than 43000 women, those who slept with a TV or room light on were more likely to gain weight over five years. Women exposed to light during sleep had a 17 percent higher chance of gaining at least 11 pounds compared to those who slept in complete darkness.

This effect remained even after accounting for diet, exercise, and total sleep time, showing that the light itself was the issue.

<https://www.nih.gov/news-events/nih-research-matters/artificial-light-during-sleep-linked-obesity>

When your body clock is out of sync due to bright nights and dim days, hormones like leptin and insulin lose their rhythm. This leads to increased fat storage and poor blood sugar control.

Artificial light at night quietly erases many of the benefits gained from eating well and getting sunlight during the day.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC4377487/>

Strategies for Health After Sunset: Circadian Rhythm and Light Exposure

When it comes to maintaining optimal health, managing light exposure after sunset is crucial. Exposure to artificial visible light at night, especially from LEDs, disrupts our natural circadian rhythms and has detrimental effects on sleep quality and overall well-being.

After-Sunset Tips:

1. **Limit Screen Time After Dark:** Avoid screen devices after sunset whenever possible. If using screens is unavoidable, always wear blue light-blocking glasses. LED screens emit high levels of blue light, mimicking natural daylight and signaling the body to stay alert. This artificial stimulation raises cortisol levels, the wakefulness hormone, and interferes with our ability to wind down, disrupting leptin's role in regulating energy availability and suppressing melatonin production.
2. **Understand Melatonin's Role:** Often called the "sleep hormone," melatonin is a powerful antioxidant and anti-cancer agent, crucial not only for restful sleep but also for metabolic health. Melatonin works in tandem with leptin, helping to signal energy availability and fat-burning processes during sleep. Even brief exposure to blue light can halt melatonin production, disrupting this hormonal balance.
3. **Preserve the Body's Natural Rhythms:** The hypothalamus, where our master clock resides, detects light signals through the eyes and controls energy regulation. In the absence of blue or bright light for about four hours after sunset, leptin initiates a cascade of hormonal responses, beginning with melatonin and thyroid hormone release. This chain reaction is essential for maintaining metabolism, releasing growth hormone, and burning fat during sleep.
4. **Minimize Artificial Light Exposure in Public:** If you must go into brightly lit areas after sunset, wear blue light-blocking glasses with dark orange or red tints, which mimic the natural light of fire and effectively block sleep-disrupting blue light. Lighter lenses often do not block sufficient blue light, potentially leaving you feeling unrested and unhealed by morning.
5. **Create a Sleep Sanctuary:** Both visible and non-visible (EMF) artificial light in the bedroom can impair sleep quality and hinder recovery. Your bedroom should be a haven for rest, with minimal artificial light sources to prevent interference with your body's natural repair processes.

Understanding Circadian Rhythm Disruption: Circadian rhythms regulate numerous physiological processes. Disruptions to this rhythm, especially from exposure to artificial blue light, late meals, and elevated stress, can impair overall health:

- ▲ **Artificial Light Exposure:** LEDs and screens at night trick the brain into thinking it's still daytime, delaying melatonin production and impacting sleep-related repair mechanisms.
- **Late Evening Meals:** Eating late signals the body to prioritize digestion, diverting energy from essential cellular and biological repair processes.
- **Evening Stress:** Elevated stress in the evening, whether from intense exercise or mental activity, raises cortisol, making it difficult for the brain to transition into a relaxed state.
- **Relying on Sleep Aids:** With the body's natural relaxation cycle disrupted, individuals may turn to prescription sleep aids or alcohol, which cannot replicate the restorative benefits of darkness.

Consequences of Circadian Rhythm Disruption: Circadian disruption significantly affects physical and emotional health. Common consequences include:

- **Poor Sleep Quality:** Difficulty falling or staying asleep, frequent awakenings, and a sense of fatigue upon waking, with diminished mood and mental clarity.
- **Long-term Health Impacts:** Chronic circadian disruption prevents proper mitochondrial repair, leading to degenerative diseases related to impaired cellular energy production. These include vision issues, autoimmune conditions, chronic fatigue, obesity, heart disease, and cancer.
- **Diminished Quality of Life:** As circadian disruption compounds, it not only affects daily well-being but also contributes to an array of modern chronic illnesses that limit vitality and life enjoyment.

Repair and recovery are most effective when aligned with our natural circadian rhythm. Protecting this rhythm by controlling light exposure after sunset is a fundamental, yet often overlooked, factor in preventing the onset of chronic illnesses that plague our modern world.

Minimizing artificial light exposure after sunset and creating an EMF-free sleep sanctuary can transform your evenings, helping you relax naturally without substances, fall asleep easily, and enjoy deep, uninterrupted sleep. As you wake up more refreshed, you'll notice increased energy, a positive outlook, greater resilience to stress, and improved focus and productivity. Over time, these changes can also alleviate physical discomfort and enhance chronic health conditions, contributing to a stronger sense of well-being.

Scientific References

Circadian Rhythms

- [Circadian rhythms, sleep, and metabolism - PMC](#)
- [Assessment of Circadian Rhythms - PMC](#)
- [The genetics of circadian rhythms, sleep and health](#)
- [Effects of Intermittent Fasting on the Circulating Levels and Circadian Rhythms of Hormones](#)
- [The genetics of circadian rhythms, sleep and health](#)
- [Effects of Intermittent Fasting on the Circulating Levels and Circadian Rhythms of Hormones - PMC](#)
- [Overview of Circadian Rhythms - PMC](#)
- [Circadian Rhythm Dysregulation and Restoration: The Role of Melatonin - PMC](#)
- [Circadian rhythms and cancers: the intrinsic links and therapeutic potentials](#)
- [Role of melatonin in the regulation of human circadian rhythms and sleep](#)
- [Circadian Rhythm and the Gut Microbiome](#)
- [Treating Circadian Rhythm Disruption in Bipolar Disorder - PMC](#)
- [Circadian rhythms and the kidney](#)

- [Circadian rhythms and pain](#)
- [Bile acid metabolism and circadian rhythms](#)
- [Circadian rhythms: influence on physiology, pharmacology, and therapeutic interventions](#)
- [Genetics of Circadian Rhythms - PMC](#)
- [Circadian Rhythms: Implications for Health and Disease](#)
- [The aging clock: circadian rhythms and later life](#)
- [Circadian Rhythms and the Transcriptional Feedback Loop \(Nobel Lecture\)*](#)
- [Circadian rhythms](#)
- [Melatonin: Physiological effects in humans](#)
- [When Rhythms Meet the Blues: Circadian Interactions with the Microbiota-Gut-Brain Axis](#)
- [Full article: Systematic review of light exposure impact on human circadian rhythm](#)
 - <https://nigms.nih.gov/education/fact-sheets/Pages/circadian-rhythms.aspx#:~:text=Circadian%20rhythms%20are%20physical%2C%20mental,the%20study%20of%20circadian%20rhythms.>

Blue Light Effects / Risks

- [Mitochondria as Potential Targets and Initiators of the Blue Light Hazard to the Retina - PMC](#)
- [The inner clock-Blue light sets the human rhythm](#)
- [Research progress about the effect and prevention of blue light on eyes - PMC](#)
- [Retinal damage induced by commercial light emitting diodes \(LEDs\) - ScienceDirect](#)
- [The Role of Oxidative Stress and Autophagy in Blue-Light-Induced Damage to the Retinal Pigment Epithelium in Zebrafish In Vitro and In Vivo](#)
- [Effects of blue light on the circadian system and eye physiology](#)
- [Blue Light Protection, Part I-Effects of blue light on the skin](#)
- [Mechanisms of blue light-induced eye hazard and protective measures: a review](#)

Near-Infrared Light

- [Near Infrared \(NIR\) Light Therapy of Eye Diseases: A Review - PMC](#)
- [Aging retinal function is improved by near infrared light \(670 nm\) that is associated with corrected mitochondrial decline - PMC](#)

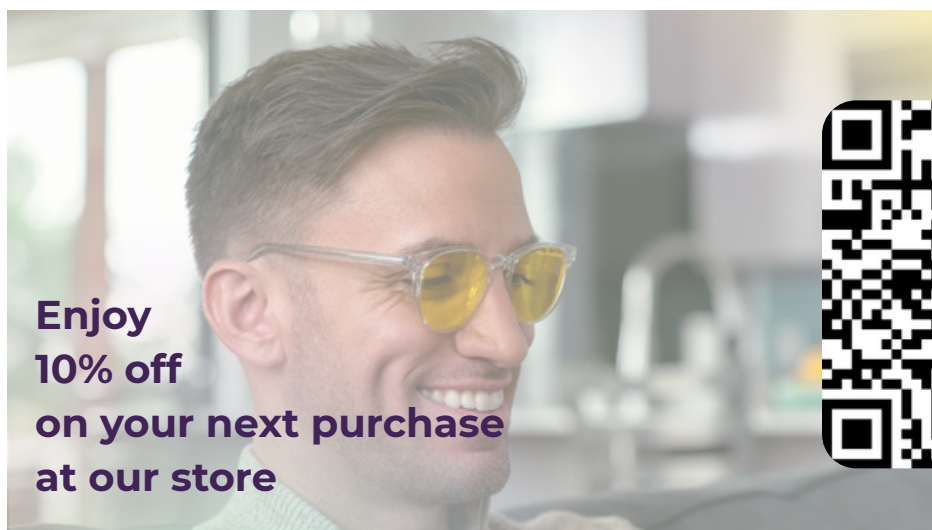
Melatonin

- [Protecting the Melatonin Rhythm through Circadian Healthy Light Exposure](#)
- [Blue light from light-emitting diodes elicits a dose-dependent suppression of melatonin in humans](#)
- [Ocular and systemic melatonin and the influence of light exposure](#)

- [Is Melatonin the “Next Vitamin D”? A Review of Emerging Science, Clinical Uses, Safety, and Dietary Supplements](#)

Studies on Blue Light Blocking Glasses

- [Blocking nocturnal blue light for insomnia: A randomized controlled trial](#)
- [A double-blind, randomized, placebo-controlled trial of adjunctive blue-blocking glasses for the treatment of sleep and circadian rhythm in patients with bipolar disorder](#)
- [Wearing blue light-blocking glasses in the evening advances circadian rhythms in the patients with delayed sleep phase disorder: An open-label trial](#)
- [Blue-Light Filtering Spectacle Lenses: Optical and Clinical Performances](#)
- [Effect of blue-blocking glasses in major depressive disorder with sleep onset insomnia: A randomized, double-blind, placebo-controlled study](#)



**Enjoy
10% off
on your next purchase
at our store**